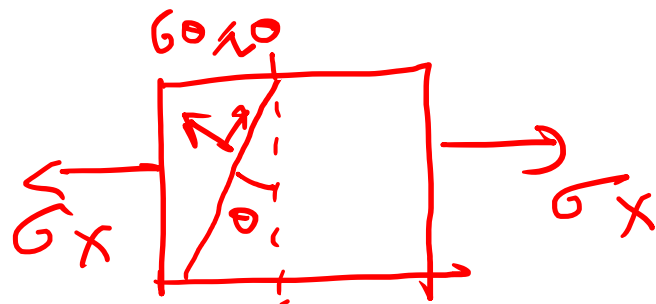


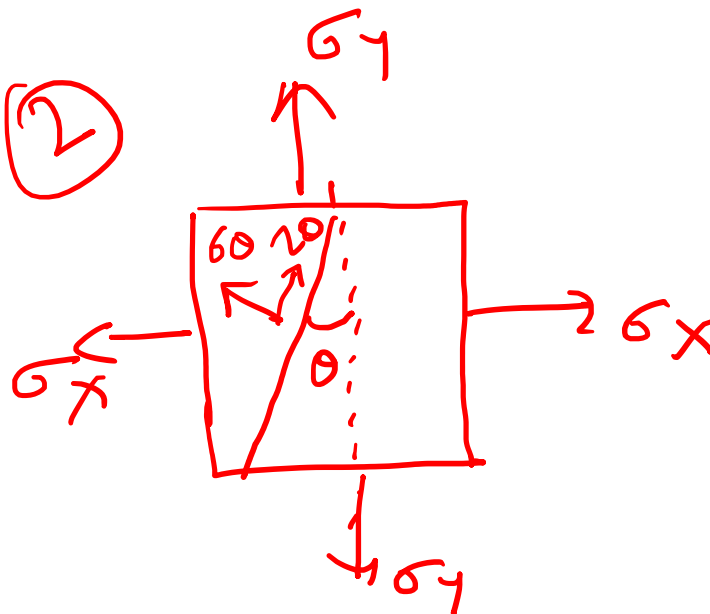
# Mechanics of Solids

①



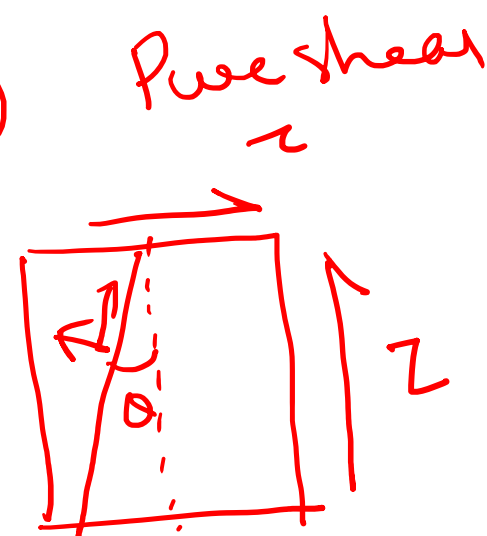
Uniaxial  
 $\sigma_0, \epsilon_0$

②



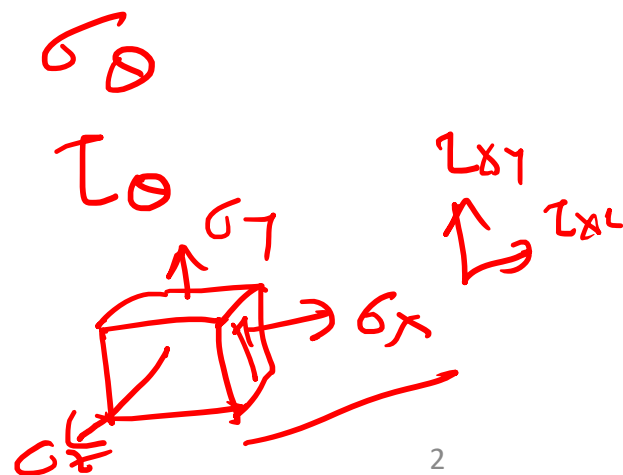
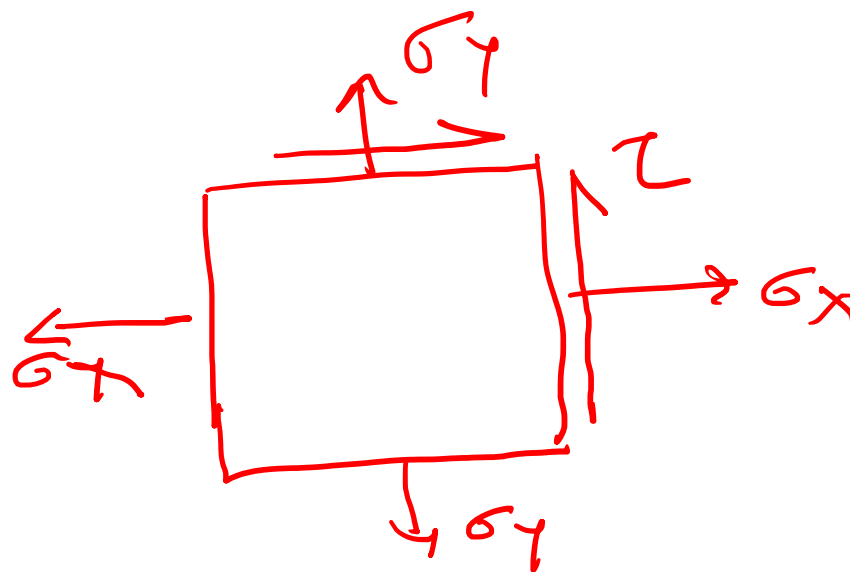
$\sigma_0, \epsilon_0$

③



$\sigma_0, \epsilon_0$

④



# Principal Stress: Mohr's Circle

**Example 2.6** || The stresses on two perpendicular planes through a point in a body are 30 MPa and 15 MPa both tensile along with a shear stress of 25 MPa. Find

- (i) the magnitude and direction of principal stresses
- (ii) the planes of maximum shear stress
- (iii) the normal and shear stresses on the planes of maximum shearing stress

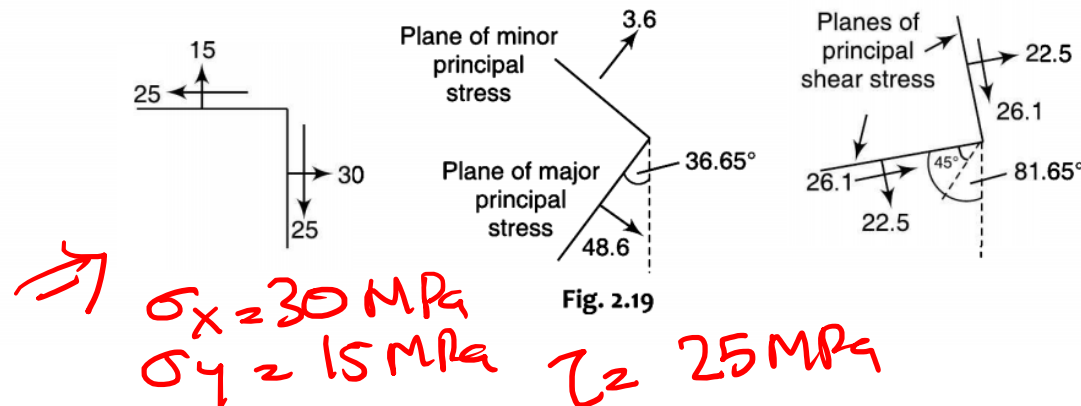
**Given** A biaxial and shear stress system,

$$\sigma_x = 30 \text{ MPa} \quad \sigma_y = 15 \text{ MPa} \quad \tau = 25 \text{ MPa}$$

**To find**

- Principal stresses
- Planes of maximum shear stresses
- Normal and shear stresses on the planes of maximum shear stress.

Refer Fig. 2.19.



### Principal stresses

$$\begin{aligned}\sigma_1, \sigma_2 &= \frac{1}{2}(\sigma_x + \sigma_y) \pm \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2} \\ &= \frac{1}{2}(30 + 15) \pm \frac{1}{2}\sqrt{(30 - 15)^2 + 4(25)^2} \\ &= 22.5 \pm \frac{1}{2}\sqrt{225 + 2500} \\ &= 22.5 \pm 26.1 \\ &= 48.6 \text{ MPa (tensile) and } -3.6 \text{ MPa (compressive)}\end{aligned}$$

$$\tan 2\theta = \frac{2\tau}{\sigma_x - \sigma_y} = \frac{2 \times 25}{30 - 15} = \frac{50}{15} = 3.333$$
$$2\theta = 73.3^\circ$$

or

or

$$\theta_1 = 36.65^\circ \quad \text{and} \quad \theta_2 = 36.65^\circ + 90^\circ = 126.65^\circ$$

The major principal stress of 48.6 MPa occurs on the plane inclined at  $36.65^\circ$  with the plane of 30 MPa tensile stress (clockwise), and the minor principal stress occurs on plane at  $126.65^\circ$ . This can be verified from the following calculation:

$$\sigma = 30 \cos^2 36.65^\circ + 15 \sin^2 36.65^\circ + 25 \sin [2(36.65^\circ)] = 48.6 \text{ MPa}$$

### Planes of maximum shear stresses

Maximum shear stress occurs on planes at  $45^\circ$  to the principal planes, i.e., on planes at  $36.65^\circ + 45^\circ = 81.65^\circ$  and  $126.65^\circ + 45^\circ = 171.65^\circ$  (or at  $36.65^\circ - 45^\circ = -8.35^\circ$  which is the same plane as  $171.65^\circ$ )

### Stresses on planes of maximum shear stress

Normal stress on planes of maximum shear stress,

$$\begin{aligned}\sigma_{81.65} &= \sigma_x \cos^2 \theta + \sigma_y \sin^2 \theta + \tau \sin 2\theta \\ &= 30 \cos^2 81.65^\circ + 15 \sin^2 81.65^\circ + \tau \sin 2(81.65^\circ) \\ &= 0.632 + 14.684 + 7.184 \\ &= 22.5 \text{ MPa}\end{aligned}$$

On the plane at  $-8.35^\circ$ , the stress value will be found to be same.

The above result is also obtained from,

$$\sigma = \pm \frac{1}{2}(\sigma_x + \sigma_y) = \pm \frac{1}{2}(30 + 15) = \pm 22.5 \text{ MPa} \quad (\text{Eq. 2.26a})$$

Shear stress on planes of maximum shear stress,

$$\tau_{\max} = \pm \frac{1}{2}(48.6 - (-3.6)) = \pm 26.1 \text{ MPa}$$

To correlate the direction of the maximum shear stress, it is necessary to calculate the shear stress at one of the angles,

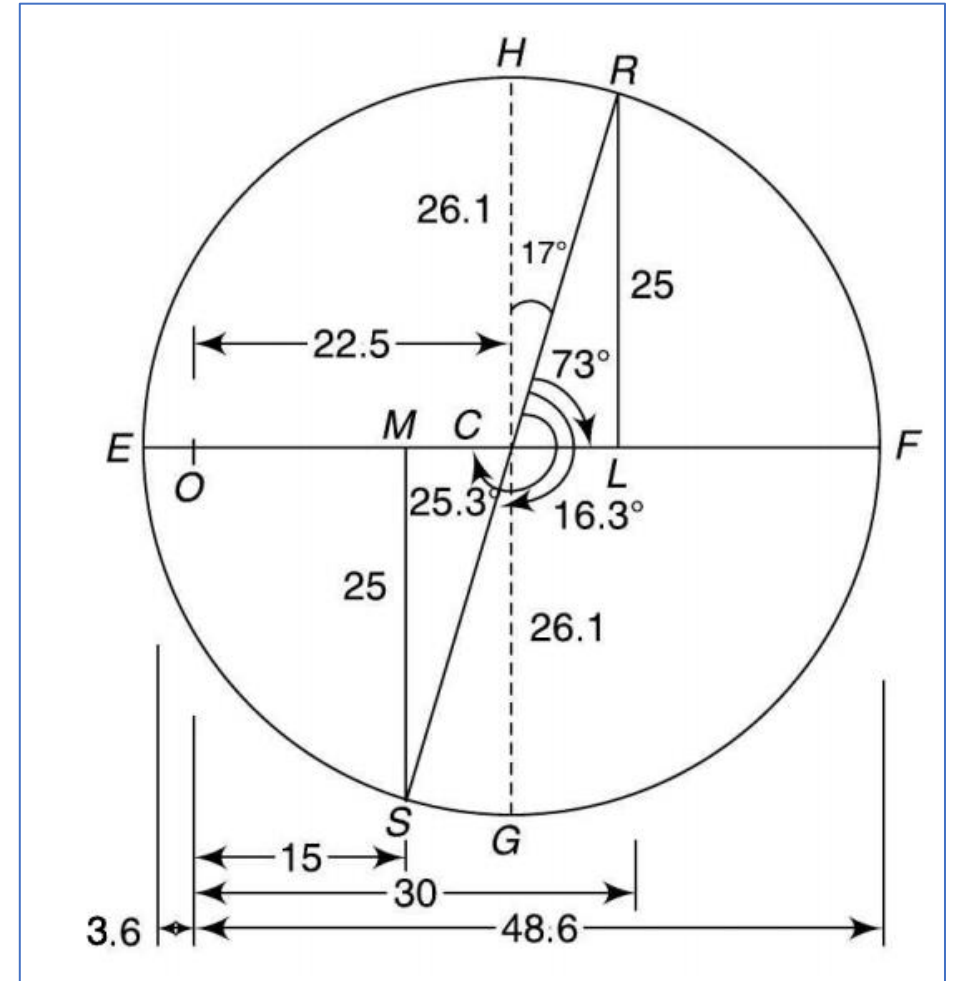
$$\begin{aligned}\tau_{81.65} &= -\frac{1}{2}(30 - 15) \sin 2(81.65^\circ) + 25 \cos 2(81.65^\circ) \\ &= -2.15 - 23.95 = -26.1 \text{ MPa}\end{aligned}$$

Thus, on a plane inclined at  $81.65^\circ$  with the plane of 30 MPa, the shear stress is 26.1 MPa counter-clockwise and on a plane inclined at  $171.65^\circ$ , the shear stress is 26.1 MPa clockwise.

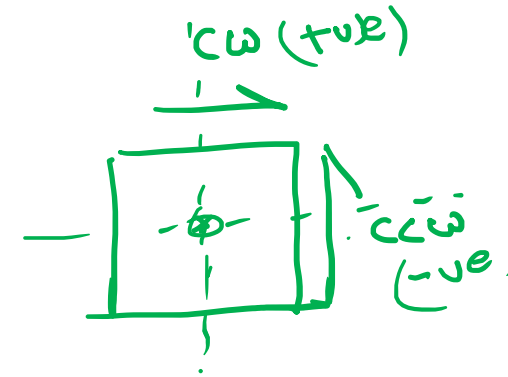
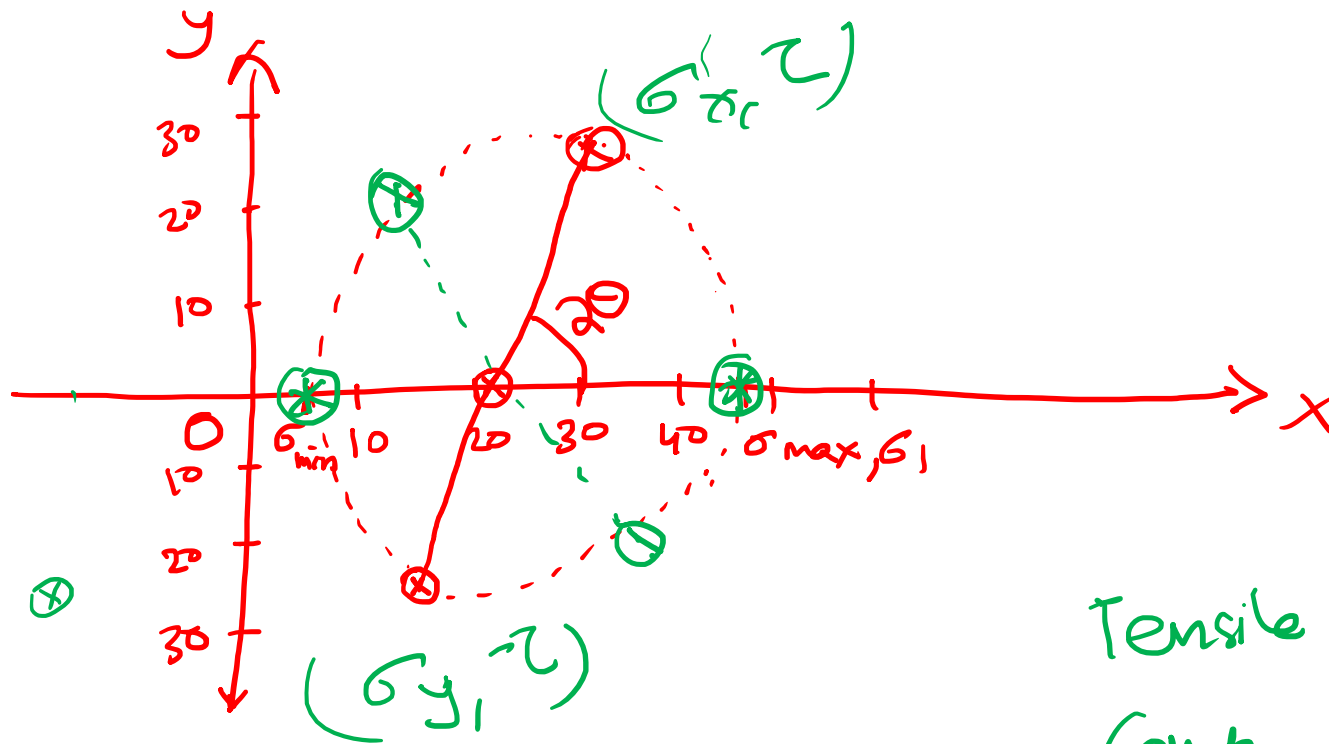
### Solution by Mohr's circle

Solution by Mohr's circle is shown in Fig. 2.20 which is self-explanatory.

# Mohr's Circle



# Mohr's Circle



$$\Rightarrow \sigma_x = 30 \text{ MPa}$$

$$\Rightarrow \sigma_y = 15 \text{ MPa}$$

$$\Rightarrow \tau = 25 \text{ MPa}$$

$$(\sigma_x, \tau)$$

$$(\sigma_y, -\tau)$$

Tensile (+ve)  
Comp (-ve)

$$\sigma_1 > \sigma_2$$

$$\sigma_1 \Rightarrow \sigma_{\max}$$

$$\sigma_2 \Rightarrow \sigma_{\min}$$

$$\theta = \frac{2\phi}{\tau}$$

In brief,  $OL = 30$  MPa,  $OM = 15$  MPa,  $LR = MS = 25$  MPa. Mohr's circle is drawn with  $C$ , the midpoint of  $LM$  as centre and passing through  $R$  and  $S$ .

- Major principal stress =  $OF = 48.6$  MPa (tensile) at angle  $73/2 = 36.5^\circ$  clockwise of  $CR$  or plane of 30 MPa tensile stress.
- Minor principal stress =  $OE = 3.6$  MPa (compressive) at angle  $253/2 = 126.5^\circ$  clockwise of  $CR$  or plane of 30 MPa tensile stress.

Maximum shear stress =  $CG = CH = 26.1$  MPa

- Inclination of the planes of maximum shear stresses:  
 $163/2 = 81.5^\circ$  clockwise of  $CR$  or plane of 30 MPa tensile stress and  
 $(343/2 = 171.5^\circ)$  clockwise or  $17/2 = 8.5^\circ$  counter-clockwise of  $CR$  or plane of 30 MPa tensile stress.

$$\sigma_{81.65} = OC = 22.5 \text{ MPa}$$

# Bending of Beams

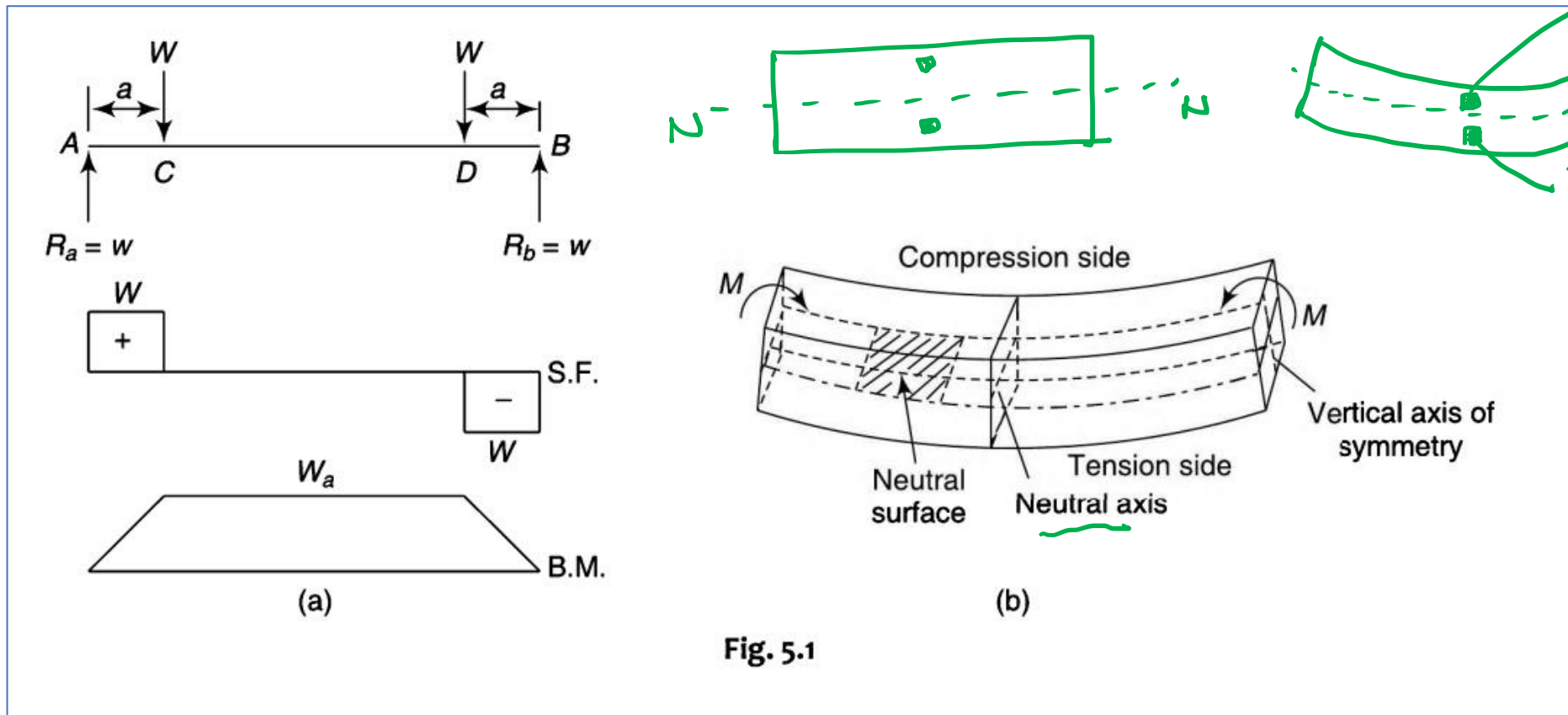


Fig. 5.1



# Bending of Beams

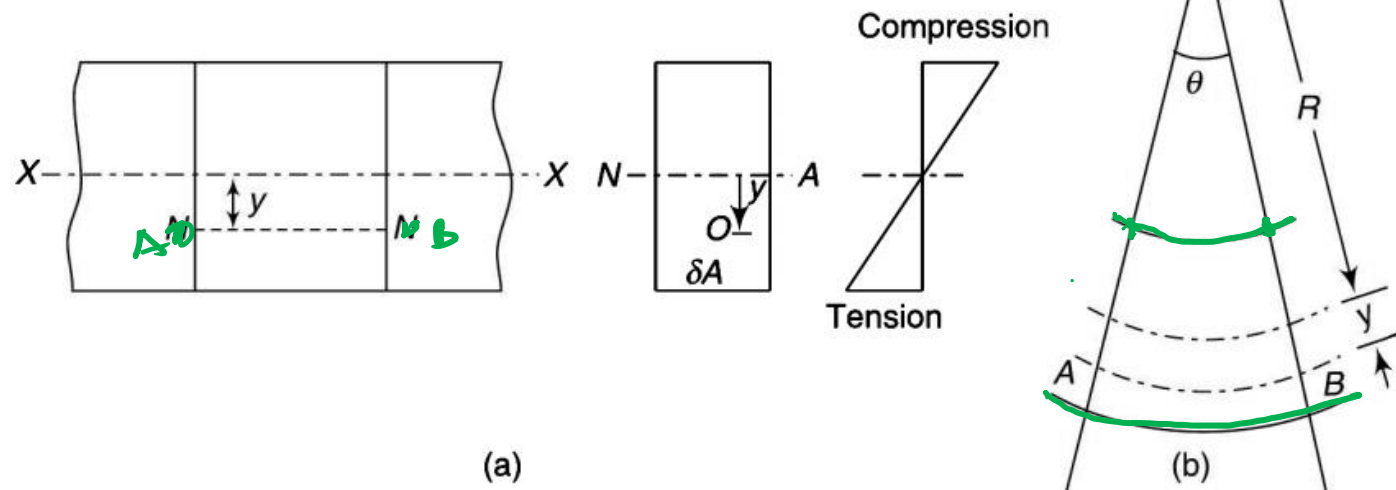
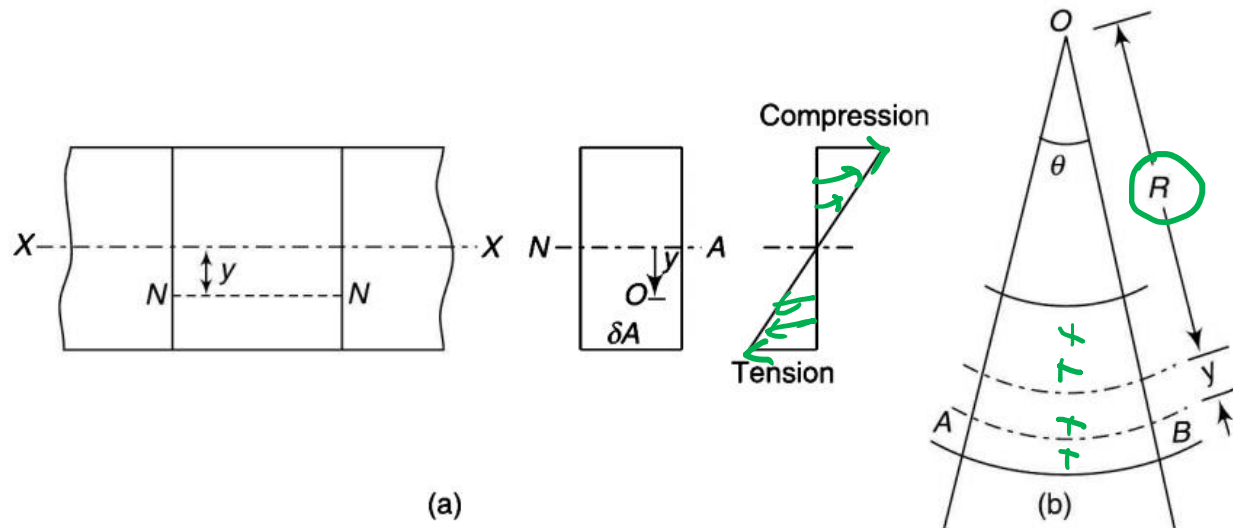


Fig. 5.2

$$l = R\theta$$

# Bending of Beams:- Pure Bending



Let  $R$  = radius of curvature of the neutral surface  
 $\theta$  = angle subtended by the beam length at centre  $O$   
 $\sigma$  = longitudinal stress

A filament of original length  $NN'$  at a distance  $y$  from the neutral axis will be elongated to a length  $AB$

The strain in  $AB = \frac{AB - NN}{NN}$  *Change in length / original length* (original length of filament  $AB$  is  $NN$ )

or  $\frac{\sigma}{E} = \frac{(R + y)\theta - R\theta}{R\theta} = \frac{y}{R}$  ✓

or  $\frac{\sigma}{y} = \frac{E}{R}$  ✓ (5.1)

Also  $\sigma = y \frac{E}{R} \propto y$  (As  $E/R$  is constant)

*Strain =  $\epsilon$*   
 *$\frac{\sigma}{E} = \text{strain}$*

Thus stress is proportional to the distance from the neutral axis  $NA$ . This suggests that for the sake of weight reduction and economy, it is always advisable to make the cross-section of beams such that most of the material is concentrated at the greatest distance from the neutral axis. Thus there is universal adoption of the I-section for steel beams.

Now let  $\delta A$  be an element of cross-sectional area of a transverse plane at a distance  $y$  from the neutral axis  $NA$  (Fig. 5.2).

For pure bending,

Net normal force on the cross-section = 0 ✓

or

$$\int \sigma \cdot dA = 0$$

or

$$\int \frac{E}{R} y \cdot dA = 0 \text{ or } \frac{E}{R} \int y \cdot dA = 0$$

or

$$\int y \cdot dA = 0$$

(5.2)

This indicates the condition that the neutral axis passes through the centroid of the section.

Also, bending moment = moment of the normal forces about neutral axis

or

$$M = \int (\sigma \cdot dA) y = \int \frac{E}{R} y \cdot dA \cdot y = \frac{E}{R} \int y^2 \cdot dA$$

$$= \frac{EI}{R}$$

(5.3)

where  $I = \int y^2 \cdot dA$  and is known as the moment of inertia or *second moment of area* of the section.

From Eqs. 5.1, 5.2 and 5.3,

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

(5.4)

# Example

**Example 5.1** || Figure 5.8 shows a simply supported 200-mm wide, 300-mm deep and 4-m long beam. Determine the bending stress at the point C which is 60 mm below the top surface and 1.2 m from the left support.

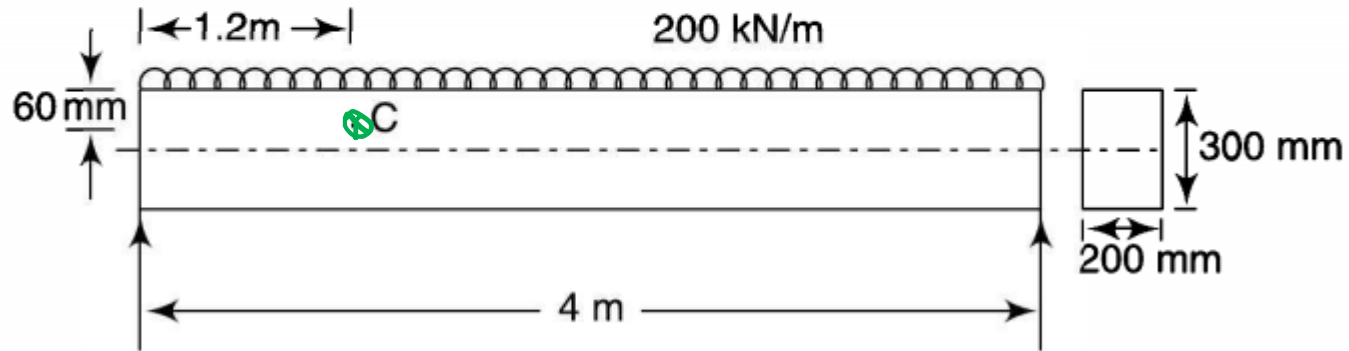
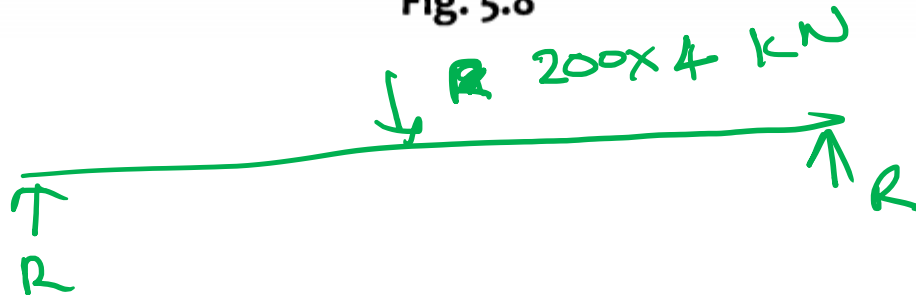


Fig. 5.8



$$\sigma = ?$$

$$\frac{M}{I} = \frac{E \cdot \kappa}{R} = \frac{\sigma}{y}$$

$$R = \frac{800}{2} = \underline{\underline{400 \text{ kN}}}$$

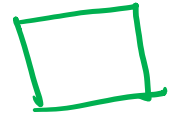
**Given** A simply supported beam

$$b = 200 \text{ mm} \quad \checkmark$$

$$d = 300 \text{ mm} \quad \checkmark$$

$$w = 200 \text{ kN/m} \quad \checkmark$$

$$L = 4 \text{ m} \quad \checkmark$$



**To find** Bending stress at the point C, 60 mm below top surface and 1.2 m from left support

**Calculation of bending moment**

$$\text{Reaction at each support, } R = \frac{200 \times 4}{2} = 400 \text{ kN} \quad \checkmark$$

$$\text{Moment of inertia of the section} = \frac{200 \times 300^3}{12} = 450 \times 10^6 \text{ mm}^4 \quad \checkmark$$

$$\text{Bending moment at cross-section at } C = R \cdot x - \frac{wx^2}{2} = 400 \times 1.2 - \frac{200 \times 1.2^2}{2} = 336 \text{ kN}\cdot\text{m}$$

Handwritten notes and diagram:

$$\frac{M}{I} = \frac{\sigma}{R} = \frac{\sigma}{y} \quad \checkmark$$



**Calculation of bending stress**

Distance of C from the neutral axis =  $150 - 60 = 90 \text{ mm}$

$$\text{Now, } \frac{\sigma}{y} = \frac{M}{I} \quad \text{or} \quad \sigma = \frac{336 \times 10^6}{450 \times 10^6} \times 90 = 67.2 \text{ MPa} \quad \checkmark$$

# Theories of failure

- A material is considered failed, when a permanent/non recoverable deformation occurs
- It can be direct separation of particles (brittle) or slipping of particles (ductile)
- Machine components and structural members are generally designed on the hypothesis that material will not yield during expected loading conditions
- Ex: for a uniaxial loading in ductile material, stress produced by load is less than the yield stress of material
- In multiaxial loading, above mentioned criterial is not sufficient

# Theories of failure

- Criteria's used for various load conditions and materials are known as Theories of Failure
  - Theories of failure
    - Max. principal stress theory – Rankine
    - Max. principal strain theory – St. Venants
    - Max. strain energy – Beltrami
    - Distortional energy – von Mises
    - Max. shear stress theory – Tresca
    - Octahedral shear stress theory

# 1. Rankine theory

- Also known as maximum principal stress theory
- Failure of material will occur when the max. principal stress in complex system attains the value of max. stress at the elastic limit in simple tension or minimum principal stress attains the elastic limit in simple compression

Let  $\sigma_x, \sigma_y, \tau$  = Direct and shear stresses on given planes in the complex system

and  $\sigma_1$  = Maximum principal stress

$$\text{Then } \sigma_1 = \frac{1}{2}(\sigma_y + \sigma_x) + \frac{1}{2}\sqrt{(\sigma_y - \sigma_x)^2 + 4\tau^2}$$

It should not be more than maximum stress in simple tension  $\sigma$ , or in the limit,  $\sigma_1 = \sigma$

In a similar way, the minimum principal stress (numerical value) should not be more than maximum stress in simple compression.

This theory is found to give good results when applied to brittle materials such as cast iron.



## 2. Guest's and Tresca's Theory

- Also known as maximum shear stress theory
- Failure occurs when the max. principal shear stress in complex system attains the value of max. shear stress at elastic limit in simple tension
- This theory is preferred in case of ductile materials such as mild steel

$$\tau = \frac{1}{2}(\sigma_1 - \sigma_2) = \frac{1}{2}\sqrt{(\sigma_y - \sigma_x)^2 + 4\tau^2}$$

### 3. St. Venant's Theory

- Also known as maximum principal strain theory
- According to this theory, max. principal strain in the complex stress system must be less than the elastic limit in simple tension if there is to be no failure

$$\varepsilon_1 = \left( \frac{\sigma_1 - \nu\sigma_2 - \nu\sigma_3}{E} \right) = \frac{\sigma}{E}$$

$$\sigma_1 - \nu\sigma_2 - \nu\sigma_3 = \sigma$$

## 4. Haigh's theory

- Also known as maximum strain energy theory
- Mostly used for ductile materials
- Work done in bringing a body to a particular state is independent of the method applied to bring the body to that state
- Failure takes place when strain energy per unit volume of body reaches the value of strain energy at elastic limit in simple tension

$$\begin{matrix} \sigma_1 & \sigma_2 & \sigma_3 \\ \sigma_4 & \sigma_5 & \sigma_6 \\ \sigma_7 & \sigma_8 & \sigma_9 \end{matrix}$$

$$\frac{1}{2E}(\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)) = \frac{\sigma^2}{2E}$$

$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) = \sigma^2$$

## 5. Mises and Henkely's theory

- Also known as maximum shear energy theory
- Failure takes place when shear strain energy in a complex system becomes equal to that in simple tension
- Applications: Ductile materials

Shear strain energy in simple tension

$$= \frac{1}{12G} [(\sigma - 0)^2 + (0 - 0)^2 + (0 - \sigma)^2] = \frac{2\sigma^2}{12G} = \frac{\sigma^2}{6G}$$

Therefore, in the limit

$$[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] = 2\sigma^2$$

# Example-1

**Example 15.1** || Principal stresses in a cast-iron body are 40 MPa tensile and 90 MPa compressive, third principal stress being zero. Determine the factor of safety based on elastic limit, if the criterion of failure is principal stress theory. The elastic limit in simple tension is 80 MPa and in simple compression 450 MPa for cast iron.

**Solution**

**Given**

Principal stresses:  $\sigma_t = 40$  MPa ✓

$\sigma_c = 90$  MPa ✓

Elastic limits:  $\sigma_t = 80$  MPa ✓

$\sigma_c = 450$  MPa ✓

**To find** Factor of safety

**Maximum principal stress criterion**

Failure takes place when the maximum principal tensile stress reaches the value of maximum stress at the elastic limit.

Maximum principal stress = 40 MPa

Elastic limit = 80 MPa

Factor of safety =  $80/40 = 2$  ✓

*Tensile*

**Minimum principal stress criterion**

Failure takes place when the minimum principal stress (maximum compressive stress) reaches the value of maximum compressive stress at the elastic limit.

Minimum principal stress,  $\sigma = 90$  MPa

Elastic limit = 450 MPa

Factor of safety =  $450/90 = 5$  ✓

Thus the failure of the material will occur due to tensile principal stress.

∴ factor of safety = 2 ✓

$\sigma_1 = 40 \text{ MPa (tensile)}$   
 $\sigma_2 = 90 \text{ MPa (compressive)}$   
 $\sigma_3 = 0 \text{ MPa}$   
 $FS = ?$

# Example-2

$$\sigma_1 = 50 \text{ MPa (T)} \\ \sigma_2 = 95 \text{ MPa (C)} \quad \left( \begin{array}{l} \text{Tensile} \\ (+ve) \end{array} \right)$$

**Example 15.2** || Principal stresses in a mild steel body are 50 MPa tensile and 95 MPa compressive, third principal stress being zero. Determine the factor of safety based on elastic limit, if the criterion of failure is maximum shear stress theory. The elastic limit in simple tension as well as in compression is 210 MPa. What will be the factor safety if the stress with 95 MPa is also tensile instead of compressive?

**Solution**

**Given**

Principal stresses:  $\sigma_t = 50 \text{ MPa}$  ✓  $\sigma_c = 95 \text{ MPa}$  ✓  
Elastic limits:  $\sigma_t = \sigma_c = 210 \text{ MPa}$  ✓

**To find**

- Factor of safety
- Factor of safety if 95 MPa stress is also tensile

***With stresses of opposite sign***

In maximum shear stress theory with stresses of opposite sign,

$$\text{Factor of safety} = \frac{\sigma}{\sigma_1 - \sigma_2} = \frac{210}{50 - (-90)} = \frac{210}{140} = 1.5$$

***With stresses of same sign***

In maximum shear stress theory with stresses of same sign,

$$\text{Factor of safety} = \frac{\sigma}{\sigma_1} \quad \text{or} \quad \frac{-\sigma}{-\sigma_2}$$

i.e.  $\frac{210}{50}$  or  $\frac{-210}{-90}$  i.e. 4.2 or 2.33

Thus Factor of safety is 2.33 in this case. ✓

Thank You